

Question				Marking details	Marks Available													
					AO1	AO2	AO3	Total	Maths	Prac								
4	(a)			A – palisade mesophyll/ parenchyma (1) {Vertical/ elongated/ long/ columnar} cells/ high number of chloroplasts (1) B – spongy mesophyll (1) Air {spaces/ pockets}/ large surface area (1)	4			4										
	(b)	(i)		Any 2 (x1) from Humidity (1) Air currents/ wind speed/ air movement (1) Light <u>intensity</u> / wavelength of light(1)	2			2		2								
		(ii)		Any one from $\text{mm}^3 \text{cm}^{-2} \text{min}^{-1}$ $\text{mm}^3 \text{min}^{-1} \text{cm}^{-2}$ $\text{mm}^3/\text{cm}^2/\text{min}$ $\text{mm}^3/\text{min}/\text{cm}^2$		1		1	1	1								
		(iii)		{gap/ difference} between {range/ error} bars/ range bars do not overlap (1)			1	1	1	1								
		(iv)		Any <u>suitable</u> xeromorphic adaptation (1) correct explanation (1) <table><tr><td>Adaptation</td><td>Explanation</td></tr><tr><td>Fewer stomata</td><td>Fewer spaces through which water vapour can be lost</td></tr><tr><td>Thick cuticle</td><td>Waterproof/ reduce evaporation</td></tr><tr><td>Hairs/ sunken stomata/ stomata in pits/ rolled (curled) leaves</td><td>Trap {humid /moist} air/ trap water vapour}reduce {water potential/ diffusion gradient}</td></tr></table>	Adaptation	Explanation	Fewer stomata	Fewer spaces through which water vapour can be lost	Thick cuticle	Waterproof/ reduce evaporation	Hairs/ sunken stomata/ stomata in pits/ rolled (curled) leaves	Trap {humid /moist} air/ trap water vapour}reduce {water potential/ diffusion gradient}		2		2		
Adaptation	Explanation																	
Fewer stomata	Fewer spaces through which water vapour can be lost																	
Thick cuticle	Waterproof/ reduce evaporation																	
Hairs/ sunken stomata/ stomata in pits/ rolled (curled) leaves	Trap {humid /moist} air/ trap water vapour}reduce {water potential/ diffusion gradient}																	
		(v)		Answer= $\pm 2\%$ = 2 marks Accept $\pm 4\%$ = 2 marks (error at both ends of distance) If answer incorrect allow $1 / 50 \times 100 = 1$ mark $100/ 50= 1\text{mark}$		2		2	2	2								
		(vi)		Time over a longer distance / greater than 50 mm (1)			1	1		1								
				Question 4 total	6	5	2	13	4	7								